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A Note on Computing Opportunity Cost

You are stuck on an island. You spend the 12 hours of daylight gathering coconuts and catching fish. It takes you **1 hour** to get a coconut and **2 hours** to catch a fish.

We can represent your costs this way:

	Coconut	Fish
Time Cost	1 hour	2 hours

Q. What is the *opportunity* cost of getting a coconut? Of catching a fish?

A. By spending 1 hour getting a coconut you are giving up catching 0.5 fish. By spending 2 hours catching a fish, you give up collecting 2 coconuts. We can update the table to show this information.

	Coconut	Fish
Time Cost	1 hour	2 hours
Opportunity Cost	0.5 fish	2 coconuts

Now the same problem posed differently. Rather than writing your time costs we can write your productivity—how many goods you produce in 12 hours:

	Coconut	Fish
Quantity (in 12 hours)	12 coconuts	6 fish

Q. What is the opportunity cost of getting a coconut? Of catching a fish?

A. By getting a coconut you are giving up 0.5 fish. You can see that because 12 coconuts “cost” you 6 fish: 12 coconuts per 6 fish is the same as 1 coconut for 0.5 fish.

Similarly, by catching a fish, you give up collecting 2 coconuts.

We can update the table. Of course, the opportunity costs are the same as computed previously.

	Coconut	Fish
Quantity (in 12 hours)	12 coconuts	6 fish
Opportunity cost	0.5 fish	2 coconuts

Q. What's going on here?

A. In one case we looked at the time it takes you to "produce" a good and computed the opportunity cost from that. In the other case we looked at how many units of the good you could produce in a set amount of time. It is the same information, just presented differently.

Q. How am I to keep this straight?

A. It's all completely logical. ☺

A2. But here's a useful method. Write out fractions, paying careful attention to units. The key insight is that the opportunity cost of getting a coconut is asking "How many *fish per coconut*, i.e., $\frac{\text{fish}}{\text{coconut}}$, do I give up?" It's like saying "What is the price of a coconut in terms of fish?" You are used to asking for price in terms of \$\$, i.e., $\frac{\$}{\text{coconut}}$. This is no different!

By remembering you want $\frac{\text{fish}}{\text{coconut}}$ and paying careful attention to units, you can't go wrong! Remember, for the opportunity cost you want fish/coconut. So you have to divide in the right way to get that. For example:

	Coconut	Fish
Time Cost	1 hour/coconut	2 hours/fish
Opportunity Cost	$\frac{1 \frac{\text{hour}}{\text{coconut}}}{2 \frac{\text{hour}}{\text{fish}}} = \frac{1}{2} \frac{\frac{1}{\text{coconut}}}{\frac{1}{\text{fish}}} = \frac{1}{2} \frac{\text{fish}}{\text{coconut}}$	

or

	Coconut	Fish
Quantity (in 12 hours)	12 coconuts/day	6 fish/day
Opportunity Cost	$\frac{6 \frac{\text{fish}}{\text{day}}}{12 \frac{\text{coconuts}}{\text{day}}} = \frac{1}{2} \frac{\frac{\text{fish}}{1}}{\frac{\text{coconut}}{1}} = \frac{1}{2} \frac{\text{fish}}{\text{coconut}}$	

Now you try it for computing the opportunity cost of fish!